

YOUNGSANG KWON, Ph.D.

Associate Professor

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EDUCATION

Ph.D., Geography, State University of New York at Buffalo, Buffalo, NY	2012
M.S., Environment Studies, Seoul National University, Seoul, Korea	2004
B.S., Forest Resources and Environmental Science, Korea University, Seoul, Korea	2001

PROFESSIONAL EMPLOYMENT

Associate Professor, Department of Earth Sciences, University of Memphis	2020–Present
Assistant Professor, Department of Earth Sciences, University of Memphis	2013–2020
Visiting Assistant Professor, Department of Geography, University of Delaware	2011–2013

RESEARCH INTERESTS

Biogeography; geospatial science and GIS; remote sensing; species distribution modeling; machine learning and AI applications in environmental science; forest ecology; groundwater monitoring; urban heat and environmental justice; geoscience education; community-engaged geospatial research.

PEER-REVIEWED PUBLICATIONS

Google Scholar metrics (as of March 2026): Total citations: 484 | h-index: 13 | i10-index: 14

In Press

Van Arsdale, R., Kwon, Y., Cox, R., & Lumsden, D. (in press). Back to the future – the Mississippi River. *American Scientist*.

Under Review

Kim, Y., Hwang, I., Song, Y., Suh, J.-H., Kil, H., Kwon, Y., & Jeong, S. (under review). Evaluating GPS fix-interval sensitivity in wild boar movement analysis using BBMM-derived spatial metrics. *Ecology and Evolution*.

Kim, H.-S., Jiang, L., Wei, N., Xu, X., Wu, C., Weng, E., Kwon, Y., Liu, M., Huang, W., Zhou, Y., & Luo, Y. (under review). Disturbances halve carbon storage potential of U.S. forests. *Nature*.

Douchinsky, P. R., Crupi, A. P., Kwon, Y., Waite, J. N., & Puckett, E. E. (under major revision). Detecting changes in distribution and abundance using harvest data in two sympatric bear species in Southeast Alaska. *Journal of Environmental Management*.

Jazaei, F., Haq, F. U., Arshad, A., Babakhani, P., Waldron, B., Kwon, Y., Nazari, R., & Aslam, M. F. (under review). GRACE in academia vs. practice: A review of satellite-based groundwater monitoring and its limited uptake by local authorities and the private sector. *Journal of Environmental Management*.

Kohlmeier, P., Tickles, A., Yancey, A., Taylor, K., Abbott, A., Conley, M., Johnson, L., Coats, E., Ferguson, A., Christian, S., Kwon, Y., & Kohlmeier, P. (under review). Socioeconomic and environmental variation predicts insect reproductive fitness in urban residential areas. *Frontiers in Ecology and Evolution*.

Published

- Lee, S., Cho, C. S., Kim, Y., Kwon, Y., & Choi, E. (2026). Upper mantle heterogeneity and weak subduction boundaries control crustal stress in the Korean Peninsula. *Geophysical Research Letters*, 53(5), e2025GL119915.
- Kwon, Y., Choi, J., & Patton, M. (2025). Assessing tree migration potential: Growth deviations and range dynamics in eastern US forests. *Forest Ecology and Management*, 586, 122718.
- Sherman, S. P., III, Parish, R. M., Kwon, Y., Meredith, S., & Johnson, D. (2024). Non-destructively characterizing sandstones, orthoquartzites, agates, and petrified wood for provenance research: Perspectives from the Southeastern Coastal Plain, United States. *Geoarchaeology*, 39(6), 628–644.
- Brugere, L., Kwon, Y., Frazier, A. E., & Kedron, P. (2023). Improved prediction of tree species richness and interpretability of environmental drivers using a machine learning approach. *Forest Ecology and Management*, 539, 120972.
- Giles, K. D., Jackson, W. T., Jr., McKay, M. P., Beebe, D. A., Larsen, D., Kwon, Y., & Shaulis, B. (2023). Sediment input, alongshore transport, and coastal mixing in the northeastern Gulf of Mexico based on detrital-zircon geochronology. *Marine and Petroleum Geology*, 148, 105997.
- Van Arsdale, R., Cox, R., Lumsden, D., & Kwon, Y. (2023). The past, present, and future Mississippi River. *The Journal of Geology*, 131(3), 187–198.
- Van Arsdale, R., & Kwon, Y. (2022). Pliocene drainage in Manitoba and northwestern Ontario, Canada. *The Journal of Geology*, 130(6), 465–474.
- Baker, B. W., & Kwon, Y. (2021). Comparisons of the spatial extent of eastern US tree species between expert-drawn Little's range map and Forest Inventory and Analysis. *The Professional Geographer*, 73(3), 413–433.
- Ron-Ferguson, N., Chin, J., & Kwon, Y. (2021). Leveraging machine learning to understand urban change with net construction. *Landscape and Urban Planning*, 216, 104239.
- Neupane, D., Kwon, Y., Risch, T. S., & Johnson, R. L. (2020). Changes in habitat suitability over a two-decade period before and after Asian elephant recolonization. *Global Ecology and Conservation*, 22, e01023.
- Borie, C., Parcero-Oubiña, C., Kwon, Y., Salazar, D., Flores, C., Olguín, L., & Andrade, P. (2019). Beyond site detection: The role of satellite remote sensing in analyzing archaeological problems. A case study in lithic resource procurement in the Atacama Desert, northern Chile. *Remote Sensing*, 11(7), 869.
- Kwon, Y., & Feng, L. (2019). Investigation of the peninsula effect using the latitudinal abundance pattern for tree species in Florida. *Diversity*, 11(2), 20.
- Kwon, Y., Doty, A., Huffman, M., Rolland, V., Istvanko, D., & Risch, T. (2019). Implications of forest management practices on sex-specific habitat use by *Nycticeius humeralis*. *Journal of Mammalogy*, 100(4), 1249–1260.
- Kwon, Y., Lee, T., Lang, A., & Burnett, D. (2019). Assessment on latitudinal tree species richness using environmental factors in the southeastern United States. *PeerJ*, 7, e6781.

- Neupane, D., Kwon, Y., Risch, T. S., Williams, A. C., & Johnson, R. L. (2019). Habitat use by Asian elephants: Context matters. *Global Ecology and Conservation*, 17, e00570.
- Shin, E., Kwon, Y., & Shaban-Nejad, A. (2019). Geo-clustered chronic affinity: Pathways from socio-economic disadvantages to health disparities. *JAMIA Open*, 2(3), ooz029.
- Behrman, C., Van Arsdale, R., Kwon, Y., Counts, R., Adkins, C., & Swann, C. (2019). Drone geologic mapping of an active sand and gravel quarry, DeSoto County, Mississippi. *Drones*, 3(3), 57.
- Kwon, Y., Larsen, C. P., & Lee, M. (2018). Tree species richness predicted using a spatial environmental model including forest area and frost frequency, eastern USA. *PLoS ONE*, 13(9), e0203881.
- Kwon, Y., & Baker, B. W. (2017). Area-based fuzzy membership forest cover comparison between MODIS NPP and Forest Inventory and Analysis (FIA) across eastern US forest. *Environmental Monitoring and Assessment*, 189(1), 19.
- Choung, S., Oh, J., Han, W. S., Chon, C. M., Kwon, Y., & Shin, W. (2016). Comparison of physicochemical properties between fine (PM_{2.5}) and coarse airborne particles at cold season in Korea. *Science of the Total Environment*, 541, 1132–1138.
- Fryxell, R. T., Moore, J. E., Collins, M. D., Kwon, Y., Jean-Philippe, S. R., Schaeffer, S. M., Grossman, A., Duggan, J., Cooley, P., Krawczak, M., & Houston, A. E. (2015). Habitat and vegetation variables are not enough when predicting tick populations in the southeastern United States. *PLoS ONE*, 10(12), e0144092.
- Kwon, Y., & Larsen, C. P. (2013). Effects of forest type and environmental factors on forest carbon use efficiency assessed using MODIS and FIA data across the eastern USA. *International Journal of Remote Sensing*, 34(23), 8425–8448.
- Kwon, Y., & Larsen, C. P. (2013). An assessment of the optimal scale for monitoring of MODIS and FIA NPP across the eastern USA. *Environmental Monitoring and Assessment*, 185(9), 7263–7277.
- Kwon, Y., & Larsen, C. P. (2012). Use of pixel- and plot-scale screening variables to validate MODIS GPP predictions with Forest Inventory and Analysis NPP measures across the eastern USA. *International Journal of Remote Sensing*, 33(19), 6122–6148.

CONFERENCE PRESENTATIONS AND INVITED TALKS

Refereed Conference Presentations

- Baer, M., Burnette, D., Lombardi, R., & Kwon, Y. (2026, March). A cool- and warm-season streamflow reconstruction for Memphis, Tennessee. Paper presented at the GSA SE-NC-SC Section Meeting, Memphis, TN.
- Williford, G. M., Kwon, Y., & Wentzloff, B. L. (2026, March). Integrating GRACE-derived groundwater storage and well-log data for U.S. aquifers. Poster presented at the GSA SE-NC-SC Section Meeting, Memphis, TN.
- Sengupta, S., Yaeger, M., Williford, G., & Kwon, Y. (2025, April). Assessing stream health in West Tennessee: Advancing scalable GIS and remote sensing techniques. Paper presented at the 34th Tennessee Water Resources Symposium, Burns, TN.
- Kwon, Y., Diko, S., Larrivee, M., & Richardson, D. (2025, November). Community heat mapping and canopy analysis to enhance understanding of urban heat in Greater Memphis. Paper presented at the SEDAG Annual Meeting, Lexington, KY.

- Lombardi, A., Burnett, J., & Kwon, Y. (2025, November). Exploring new approaches for detecting precipitation whiplash events on millennial scales in the Lower Missouri River Basin. Paper presented at the SEDAAG Annual Meeting, Lexington, KY.
- Choi, J., Kwon, Y., & Patton, M. (2025). Assessing tree migration potential: Growth deviations and range dynamics in eastern US forests. Poster presented at the MAGIC Geospatial Conference, Memphis, TN.
- Kwon, Y. (2024). Assessing the influence of prescribed burns on male wild turkey movement patterns. Paper presented at the SEDAAG Conference.
- Kwon, Y. (2024). Biogeography of tree species richness and latitudinal diversity gradient. Paper presented at the AAG Annual Meeting, Honolulu, HI.
- Lombardi, R., Kwon, Y., & Burnette, D. (2024). A millennial record of precipitation volatility along the Lower Missouri River, Central USA. Copernicus Meetings.
- Kwon, Y., Ron-Ferguson, N., & Chin, J. (2021). Data-driven development: Analyzing urban change through construction, demolition, and renovation. Paper presented at the MAGIC Conference, Memphis, TN.
- Sherman III, S.P., Parish, R.M., Kwon, Y., Carr, P.J., & Greenlee, D.M. (2021). A reevaluation of surface-collected projectile points or knives from the Poverty Point site using reflectance spectroscopy. Paper presented at the 86th SAA Annual Meeting (online).
- Kwon, Y. & Van Arsdale, R.B. (2020). Pre-glacial Pliocene drainage in Manitoba and western Ontario, Canada. Paper presented at the GSA Annual Meeting (online).
- Kwon, Y. (2019). Investigation of the peninsula effect using the latitudinal abundance pattern for tree species in Florida. Paper presented at the Ecological Society of America, Louisville, KY.
- Kwon, Y. & Baker, B.W. (2018). Comparison of spatial extent between Little's tree species ranges and Forest Inventory Analysis (FIA) in the eastern U.S. Paper presented at the AAG Annual Meeting, New Orleans, LA.
- Van Arsdale, R. & Kwon, Y. (2018). Relic Appalachian uplands along the New River–Tennessee River drainage divide. Paper presented at the GSA Annual Meeting, Phoenix, AZ.

Invited Talks

- Invited speaker, Seoul Institute, Seoul, Korea (Summer 2022)
- Invited speaker, Korea Research Institute for Human Settlements, Korea (Summer 2022)
- Invited speaker, National Institute of Ecology, Korea (Summer 2022)
- Invited speaker, Geography Department, Seoul National University, Korea (Fall 2022)
- Invited speaker, Forestry Department, Seoul National University, Korea (Fall 2022)
- Invited speaker, Geology Department, Cheonnam National University, Korea (Fall 2022)
- Invited speaker, Korean Ecological Association Conference, Korea (2022)
- Invited speaker, Korean Restoration Conference, Korea (2022)
- Invited speaker, Seoul Institute, Seoul, Korea (Summer 2018)
- Invited speaker, Korea Research Institute for Human Settlements (Summer 2018)
- Invited speaker, National Institute of Ecology, Korea (Summer 2018)

GRANTS AND FUNDED PROJECTS

External Grants (Funded)

NOAA Center for Collaborative Heat Monitoring (CCHM) (\$10,000)

Role: PI | Duration: March–December 2025

Memphis Heat Watch Project

Protect Our Aquifer (POA) GIS Groundwater Contamination Dashboard (\$10,000)

Role: PI | Duration: February–December 2025

GIS-based dashboard for groundwater and contamination data visualization

USGS EDMAP Project (No. 24-0292) (\$80,041)

Role: Lead PI | Duration: 2024–2025

Assessing Spatio-Temporal Groundwater Anomaly Mapping in US Major Aquifers

USGS 104b Water Resources Research (\$48,593)

Role: PI | Duration: 2024–2025

Assessing Stream Health in West Tennessee: Advancing Scalable GIS and Remote Sensing Techniques

USDA EDMAP (\$22,326)

Role: Lead PI | Duration: 2022–2023

Mapping Spatio-temporal Groundwater Storage Changes in Memphis Aquifer: Integration of GRACE satellite, Ground-based Estimates, and Hydrologic Modeling Approach

Tennessee Water Resources Research Center (TNWRRC) (\$43,700)

Role: Co-PI (PI: B. Waldron) | Duration: 2021–2022

Improved remote-sensing imagery techniques to identify riparian corridor characteristics in headwater streams and inventory wetlands, lakes, and permanent streams in West Tennessee

NSF GeoAllies (\$3,000–\$5,000)

Role: PI | Duration: 2021–2022

Diversity in Geoscience Field Excursion Project

Internal Grants (Funded)

- FedEx Institute of Technology, Environmental Humanities Research Cluster, \$20,000, 2024–2026 (Co-PI with A. Gaggioli)
- Faculty Research Grant, \$10,000, 2023–2024 (PI) – Delayed Tree Species Response to Climate Change: Data-driven Tree Species Interaction Model in Eastern US Forest Using FIA
- CAS Research Instrumentation and Technology Initiative (RITI), \$66,000, 2024 (Co-PI) – Portable gas exchange and chlorophyll fluorescence measurement system
- Campus Community Green Fee Fund, \$2,500 per year, 2022–2025 (PI) – Earth Day Outreach Events
- FedEx Institute of Technology, Agriculture and Food Technologies Research Cluster, \$8,000–\$10,000, 2019 (PI) – Machine learning approach for predicting biodiversity
- CAS Research Equipment Funds, \$33,000, 2019 (Co-PI with A. Mickelson) – GPS instrumentation (Trimble Geo7 & TDC100)
- FedEx Institute of Technology, \$10,000, 2017 (PI) – Drone Imaging of Active Sand and Gravel Quarries

- City of Memphis & FedEx Institute of Technology, \$10,000, 2016 (PI) – A Planning Support System for Comprehensive Planning and Zoning: A Geospatial Simulation Model
- Travel Enrichment Funds, 2019, 2024

Pending Proposals

- NASA ROSES (A8. Water Resource Management), \$1,500,000 (3 years), PI – Stormwater–Groundwater Connectivity Tool: Leveraging NASA Earth Observations for Integrated Flood and Aquifer Resilience in the Mid-South. Invited for full proposal; selected as one of ~30 nationally.
- Spencer Foundation Small Research Grant, \$49,604 (2 years), PI – What Counts as Knowledge in the Age of AI? Exploring Student Epistemology in AI-Integrated Geoscience.
- Tennessee Natural Fiber and Forestry Production Center, \$186,219 (3 years), Co-PI (PI: J. Mandel) – Strategic planning proposal to the State of Tennessee.

Submitted External Proposals (Not Funded)

- NEH, \$50,054 (1 year), Co-PI – Proposal for an environmental humanities major at the University of Memphis. (2024)
- NSF Geomorphology and Land-Use Dynamics, \$315,468 (3 years), Co-PI (PI: R. Lombardi) – Precipitation volatility and its impact on flood hazards: A long-term paleohydrologic perspective on the Lower Missouri River, USA. Co-PI: D. Burnette. (2023)
- Spencer Foundation, \$399,526 (3 years), PI – Empowering climate literacy: A place-based humanities-engaged high school geoscience program. Co-PIs: M. Khajeloo, C. Kaldon, S. Baek, A. Gaggioli. (2023)
- NEH, \$49,925 (1 year), Co-PI (PI: A. Gaggioli) – Nature's narratives: A proposal for an environmental humanities minor at the University of Memphis. (2023)
- NSF Human-Environment and Geographical Sciences, \$363,936 (3 years), PI – Integrative data-driven modeling to evaluate the impact of species interactions on tree growth and migration across environmental gradients. Co-PI: H.-S. Song (University of Nebraska–Lincoln). (2021)
- CORNET (Collaborative Research Network in Adverse Childhood Experiences), \$100,000, Co-PI (PI: A. Shaban-Nejad, UTHSC) – Knowledge-based real-time risk assessment and recommender system for screening of ACEs. Co-PI: E. Howell. (2019)
- USDA NIFA – Resilient Agroecosystems in a Changing Climate, \$115,000, PI – Improving the climate change resiliency and sustainability of family forest landowners in the US South. (2017)
- NSF Ecology and Evolution of Infectious Diseases (EEID), \$2,498,225, PI – Eco-epidemiology of Rocky Mountain Spotted fever in the eastern United States. Co-PIs: T. Fryxell, M. Collins, A. Houston, Jean-Philippe, M. Kennedy. (2015)
- Institute of Education Sciences (IES), U.S. Department of Education, \$1,147,294, Co-PI (PI: P. Pavlik) – Long-term memory for mental models. Co-PIs: M. Conley, J. Braasch. (2015)
- Silicon Mechanics Research Cluster Grant, PI – Empowering University of Memphis multi-scale multi-physics computational modeling research in science and engineering. Co-PIs: W. Alexander, E. Choi, C. Cramer, E. Daub, N. DeYonker, R. Tabor, Y. Wang. (2015)
- NSF Partnerships for International Research and Education (PIRE), Co-PI (PI: J. Mandel) – A U.S./China/Australia consortium for dynamic genomic and biophotonic imaging applications for studying linked ecological human systems. Co-PIs: H.-T. Kung, J. Cole, K. Bando. (2014)

- NASA Earth Science Applications: Water Resources, \$1,529,474, Co-PI (PI: G. Emmert) – Using remote sensing to establish hybrid experimental and modeling approaches for understanding trihalomethanes formation in drinking water distribution systems. Co-PI: P. Simone. (2014)
- NSF Geography and Spatial Sciences, \$222,271, PI – Developing a multi-scale remote sensing system to quantify the biogeochemical canopy response and impacts of insect stressors in eastern deciduous forests. Co-PIs: D. Levia, C. Siegert, J. Riggins. (2013)

TEACHING EXPERIENCE

Courses Developed

- GIS Living Lab (ESCI 7515) – Community-engaged, project-based GIS course partnering with City of Memphis Sustainability Office and The Works, Inc. (New, 2025)
- GeoAI Course – AI and machine learning for geospatial applications, part of the AI for All initiative (In development, planned Fall 2026)
- Landforms (ESCI 1020 M50) – Developed fully online M50 course; co-developed customized textbook with Pearson (2018; major revision 2024–2025)
- R Programming for Earth Sciences (Seminar, ESCI 7700) – Graduate seminar on R, Git, Quarto, and geospatial analytics (2023)
- Python GIS Programming (Advanced GIS, ESCI 4525/6525) – Restructured to Python-based GIS application format (2024)
- Introduction to GIS (ESCI 4515/6515) – Transitioned from ArcGIS to dual-platform QGIS + ArcGIS Pro with Python (2018–2024)
- Remote Sensing of Environment (ESCI 4511/6511) – Rebuilt labs with open-source QGIS + SCP plugins (2018)

Courses Taught (2013–2025)

Spring 2025	GIS Living Lab (ESCI 7515, 12 students); Landforms (ESCI 1020 M50, 40 students)
Summer 2025	Landforms (ESCI 1020 M50, 50 students)
Fall 2025	Remote Sensing (ESCI 4511, 11 students); Landforms (ESCI 1020 M50, 42 students)
Spring 2024	Geographic Information Systems (ESCI 4515/6515, 13 students); Landforms (ESCI 1020 M50, 61 students)
Summer 2024	Landforms (ESCI 1020 M50, 50 students)
Fall 2024	Biogeography (ESCI 4241, 11 students); Advanced GIS (ESCI 4525/6525, 16 students)
Spring 2023	Landforms (ESCI 1020 M50, 60 students); Geographic Information Systems (ESCI 4515/6515, 9 students)
Summer 2023	Landforms (ESCI 1020 M50, 60 students)
Fall 2023	Seminar: R Programming for Earth Sciences (ESCI 7700, 9 students); Remote Sensing (ESCI 4511/6511, 7 students)
Spring 2022	Landforms (ESCI 1020 M50, 80 students); Biogeography (ESCI 4241/6241, 6 students)
Summer 2022	Landforms (ESCI 1020 M50, 6 students)

Fall 2022	Professional Development Assignment (PDA) at Seoul National University, Korea
Spring 2021	Landforms (ESCI 1020 M50, 62 students); Geographic Information Systems (ESCI 4515/6515, 17 students)
Summer 2021	Landforms (ESCI 1020 M50, 63 students)
Fall 2021	Remote Sensing (ESCI 4511/6511, 18 students); Advanced GIS (ESCI 4525/6525, 7 students)
Spring 2020	Landforms (ESCI 1020 M50, 76 students); Biogeography (ESCI 4241/6241, 9 students)
Summer 2020	Landforms (ESCI 1020 M50, 59 students)
Fall 2020	Geographic Information Systems (ESCI 4515/6515, 18 students); Seminar: R for Spatial Analysis (ESCI 7700, 13 students)
Spring 2019	Landforms (ESCI 1020 M50, 38 students); Introduction to GIS (ESCI 4515/6515, 16 students)
Fall 2019	Landforms (ESCI 1020 M50, 39 students); Advanced GIS (ESCI 4526/6525, 10 students)
Spring 2018	Landforms (ESCI 1020 M50, 34 students); Introduction to GIS (ESCI 4515/6515, 19 students)
Fall 2018	Landforms (ESCI 1020 M50, 39 students); Remote Sensing (ESCI 4511/6511, 12 students)
2013–2017	Landforms, Introduction to GIS, Advanced GIS, Remote Sensing, Biogeography, Forest Informatics, Seminar in Earth Sciences

GRADUATE STUDENT ADVISING

Doctoral Students (as Major Advisor)

- George Williford, Ph.D. (in progress, 2024–present)
- Nathan Ron-Ferguson, Ph.D. (graduated 2020) – Currently GIS & Data Scientist, National Geospatial-Intelligence Agency
- Lian Brugere, Ph.D. (graduated 2022) – Currently Principal Data Scientist, Mueller Water Products

Master's Students (as Major Advisor)

- George Williford, M.S. (graduated 2024) – Continued to Ph.D.
- Samiksha Banjara, M.S. (graduated 2025) – GIS position at urban planning consulting company, Oxford, MS
- Md Shah Naim Hredoy, M.S. (graduated 2023) – GIS Analyst, Memphis Area Transit Authority (MATA)
- Berkay Tok, M.S. (graduated 2022) – Completed TNWRRC project
- Jiaxin Liu, M.A. (graduated 2022)
- Megan Farrell, M.S. (graduated 2022) – Planner, Shelby County Office of Sustainability and Resilience
- Michelle Field, M.S. (graduated 2019) – GIS Specialist, Smith Seckman Reid, Inc. (SSR)
- Lian Feng/Brugere, M.S. – Continued to Ph.D.

- Andrew Barnett Murphy, M.S. (graduated 2017) – INTERA Incorporated
- Alison Lang, M.S. (graduated 2016) – L3 Resources, LLC

Thesis and Dissertation Committee Member

Robert Barlow (Ph.D. candidate), Delmer Rodriguez (Ph.D. candidate), Jaslyn Herold (M.S. candidate), Max Dixon (M.S. candidate), Macy Baer (M.S. candidate), Devin Richards (M.S. candidate), Samia Noor (Ph.D., graduated 2025), Narayan Adhikari (Ph.D.), Kathryn R. Proctor (Ph.D.), Simon Sherman (Ph.D., graduated), Philip Douchinsky (M.S., Biology), Md Saddam Hossain (M.S.), Ehsan Momeni (Ph.D.), Brian Rowe (M.S.), Hunter Saunders (M.S.), Sarah Swing (M.S., Biology), Katelyn Giles (M.S.), Valarie Harrison (M.S.), Preston Bradley (M.S.), Renee Reichenbacher (M.S.), Bradly Baker (Ph.D.), Alberto Jimenez (Ph.D.)

Undergraduate Mentoring

- Brian Wentzloff – Undergraduate research assistant, EDMAP project
- Dylan Whitelaw – Established internship partnership with Protect Our Aquifer (POA)
- Junwon Choi – High school mentee; guided through peer-review publication and conference presentation

PROFESSIONAL SERVICE

Editorial Service

- Guest Editor, *Forests* (Special Issue: "Modeling of Forest Dynamics and Species Distribution"), 2024–present. Reviewed 15 manuscripts; 6 accepted under editorial decision.
- Editorial Board Member, *The Journal of Ecology and Environment*, 2025–2026.
- Reviewer, KAGES (Korea–America Association for Geospatial & Environmental Sciences) and Scholarship Committee Member.
- Student Paper Competition Reviewer, American Association of Geographers (AAG), 2022–present.
- Manuscript reviewer for peer-reviewed journals in geospatial science, ecology, and environmental research (~26 manuscripts reviewed, 2020–2026).

Conference Session Organization

- Session Organizer and Chair, "AI-Driven and Machine Learning Approaches in Geospatial Modeling for Hydrologic and Geologic Mapping," GSA SE-NC-SC Section Meeting, Memphis, TN, March 2026.

Professional Organizations and Board Memberships

- Board Member, Memphis Area Geographic Information Consortium (MAGIC), 2018–present.
- Board Member, TNView (Tennessee Geospatial Education Initiative), 2023–present.
- Member, Tennessee Geographic Information Council, 2023–present.
- Board Member, Nonconnah Creek Conservancy, 2018–present.
- Faculty Affiliate, ACRE & Institute for Intelligent Systems (IIS), University of Memphis.
- Mentor, Geosciences Education & Mentorship Support (GEMS) Program.

UNIVERSITY AND DEPARTMENT SERVICE

University-Level Service

- Committee on University Undergraduate Council (UUC), 2025–present.
- Quality Enhancement Plan (QEP) Committee, 2025–present.
- Work-in-Progress Symposium (WIPS) Undergraduate Research Judge, 2025.
- Student Research Forum Judge, University of Memphis, 2018–present.

College-Level Service

- College of Arts and Sciences (CAS) Undergraduate Curricular Council – Elected Natural Sciences Representative, 2025–present.

Department-Level Service

- Chair, Undergraduate Curriculum Committee, Earth Sciences Department.
- Chair, Undergraduate Academic Advisor, Earth Sciences Department.
- Honors Assembly & Scholarship Review Chair, Earth Sciences Department.
- Library Liaison, Earth Sciences Department.
- Department Webmaster & Email Listserv Manager.
- Discover Your Major Day presenter and recruiter.
- Sigma Gamma Epsilon (SGE) Earth Science Honors Society chapter establishment (2025).

COMMUNITY OUTREACH AND ENGAGEMENT

- Earth Day Organizer & Project-Based Learning Lead (2022–2025, four consecutive years): Coordinated large-scale K–12 outreach initiative with University Middle School, CERI graduate students, and faculty. Supported by various internal and external funds including Campus Community Green Fee Fund (2024) and Environmental Humanities Research Cluster (2025).
- K–12 Mentorship and Science Education: Mentored high school student (Junwon Choi) through peer-review publication and conference presentation; coached White Station Middle School Science Olympiad team to first place.
- Environmental Humanities Research Cluster (2024–2026, Co-PI): Interdisciplinary initiative integrating humanities and sciences for environmental challenges across 15+ faculty members.
- NASA Develop Program Partner: Provided groundwater anomaly research using GRACE satellite data (with Protect Our Aquifer, B. Waldron, and S. Schoefnacker).
- Protect Our Aquifer (POA) GIS Dashboard: Translated complex environmental datasets into accessible decision-support tools for community stakeholders. Featured in Fox13 Memphis media coverage.
- Memphis Heat Watch Project (2025): One of eight cities nationwide selected under NOAA’s Center for Collaborative Heat Monitoring; collaborated with City of Memphis, The Works, and 100+ community volunteers.
- Memphis 3.0 Comprehensive Plan – Civic Common Areas and Open Space Working Group member (2018–2019).
- STEM Curriculum Consulting, St. Paul High School.
- Senior Internship Program Advisor, Lausanne Collegiate School (2018).
- Cooper-Young Historic District Arboretum, active member (2018–2020).

- Korean Language Instructor, weekly instruction every Sunday (since 2018).
- Junior Achievement Inspire Event participant, AgriCenter (2025, as ACRE affiliate).

AWARDS, HONORS, AND FELLOWSHIPS

Professional Development Assignment (PDA), University of Memphis	Fall 2022
Visiting Scholar, Seoul National University, South Korea	
Agriculture and Food Research Fellow, FedEx Institute of Technology	2019
Drones Research Fellow, FedEx Institute of Technology	2017
Smart City Research Fellow, FedEx Institute of Technology	2016
Pixoneer Scholarship Award, KAGES	2010
NSF Research Traveling Fund, Spatial Filtering Workshop	2008

PROFESSIONAL MEMBERSHIPS

- Association of American Geographers (AAG)
- American Geophysical Union (AGU)
- Korea–America Association for Geospatial and Environmental Sciences (KAGES)
- Geological Society of America (GSA)
- Ecological Society of America (ESA)

PROFESSIONAL DEVELOPMENT

- Summer Institute Mastery Program, UM3D, University of Memphis (2020) – Online teaching pedagogy
- Grantsmanship Workshop, Division of Research and Innovation, University of Memphis (2020)
- Earth Educators’ Rendezvous, Nashville, TN (Vanderbilt University, 2019)
- STEM Funds Workshop, University of Memphis (2019)
- Publishing Pipeline Workshop, led by Dr. Furtak, University of Memphis (2019)